

CSC2062 – Assignment 1

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[Date]

Introduction

This report details the creation, feature extraction and statistical analysis of a custom dataset of 112 hand-drawn images of Living and Non-Living objects.

The primary objective of this report is to evaluate 16 structural features to identify discriminators between the two categories. Through hypothesis testing, correlation analysis and machine learning models, the report will investigate the structural dependencies within the drawings and assess the predictive power of these features.

Section 1

1.1 Dataset Creation

First, I created a balanced dataset of 112 hand-drawn images comprising of 14 unique images per class. These classes were divided evenly into ‘Living’ (cherry, banana, lemon and tree) and ‘Non-Living’(envelope, golfclub, pencil and wineglass) categories.

Images

Living Object Examples			
Cherry			
			
Banana			
			
Lemon			
			
Tree			
			

Non-Living Object Examples			
Envelope			
			
Golfclub			
			
Pencil			
			
Wineglass			
			

Image Creation

Images were drawn in Gimp 2.10 on a 45x45 pixel canvas using a 1-pixel pencil, ensuring each took up similar canvas space to minimise negligible variation. Each image included slight variation however variation in some images is negligible.

1.2: Image Threshold and Processing

Portable Greymap (pgm) images were efficiently parsed using a Python script to convert each image into CSV matrices.

The script skipped the first 3 PGM headers and applied a 128 midpoint threshold where a pixel count greater than 128 became '0' (white) and less than became '1'(black).

All 112 files were saved as comma-delimited 45x45 matrices with UTF-8 encoding using the STUDENTN_LABEL_INDEX.csv naming convention.

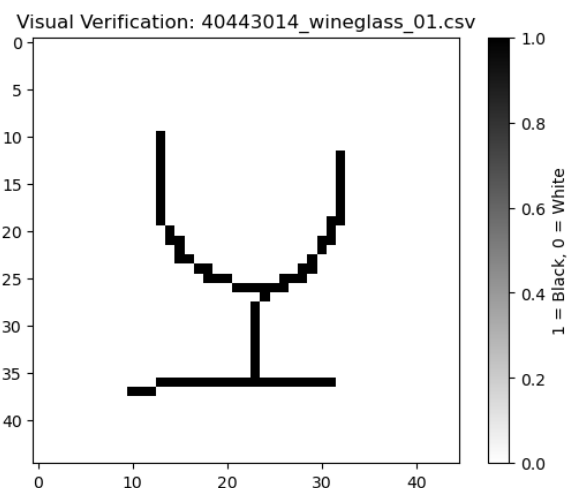
CSV Example Image

Matrix Shape: (45, 45)

[illegible]

Numeric Grid Matrix

Test block was created to verify csv file output. Using Matplotlib, mapping '1's to black and '0' to white, script was confirmed to be accurately converting files as shown in left figure.



Section 2

Task 2.1 Feature extraction

Using the numpy and scipy.ndimage libraries, 16 distinct features were extracted per image.

Custom Feature - Extent

Extent measures the area an object takes up relative to its bounding box (height * width)^[1].

While absolute limits and aspect ratio exist, they cannot differentiate objects sharing the same outside dimensions but different pixel distribution. For example, a golfclub (drawn diagonally) could end up with the exact bounding box dimensions and aspect ratio as a square envelope.

Extent was chosen as it measures how spread out the pixels are within the dimensional limits. Box like shaped objects will have a high score as the pixels will most likely trace the edges of their bounding box. In comparison, diagonal objects like the pencil and golfclub will result in a lower percentage as bounding box will consist of mainly blank space.

Extent was calculated by taking the total number of pixels (nr_pix) and dividing it by the total object area (height * width). Error handling prevented division by zero errors in the event of a blank matrix.

Task 2.2

Extracted features were appended to an array, converted to a data frame and saved to 40443014_features.csv. The dataset was sorted primarily by object label in alphabetical order and secondarily by image index.

Task 2.3

Code Explanations

All features beside connected areas, eyes and hollowness were calculated using standard NumPy matrix operations.

- **Connected Area:**

3x3 matrix^[2] was created and 'label'^[3] scanned through entire matrix to identify and number all connected pixel areas

- **Eyes:**

Matrix colour code was inverted^[4] and the 'label' function found all 'white' squares. One connected area was subtracted due to the surrounding canvas counting as a label.

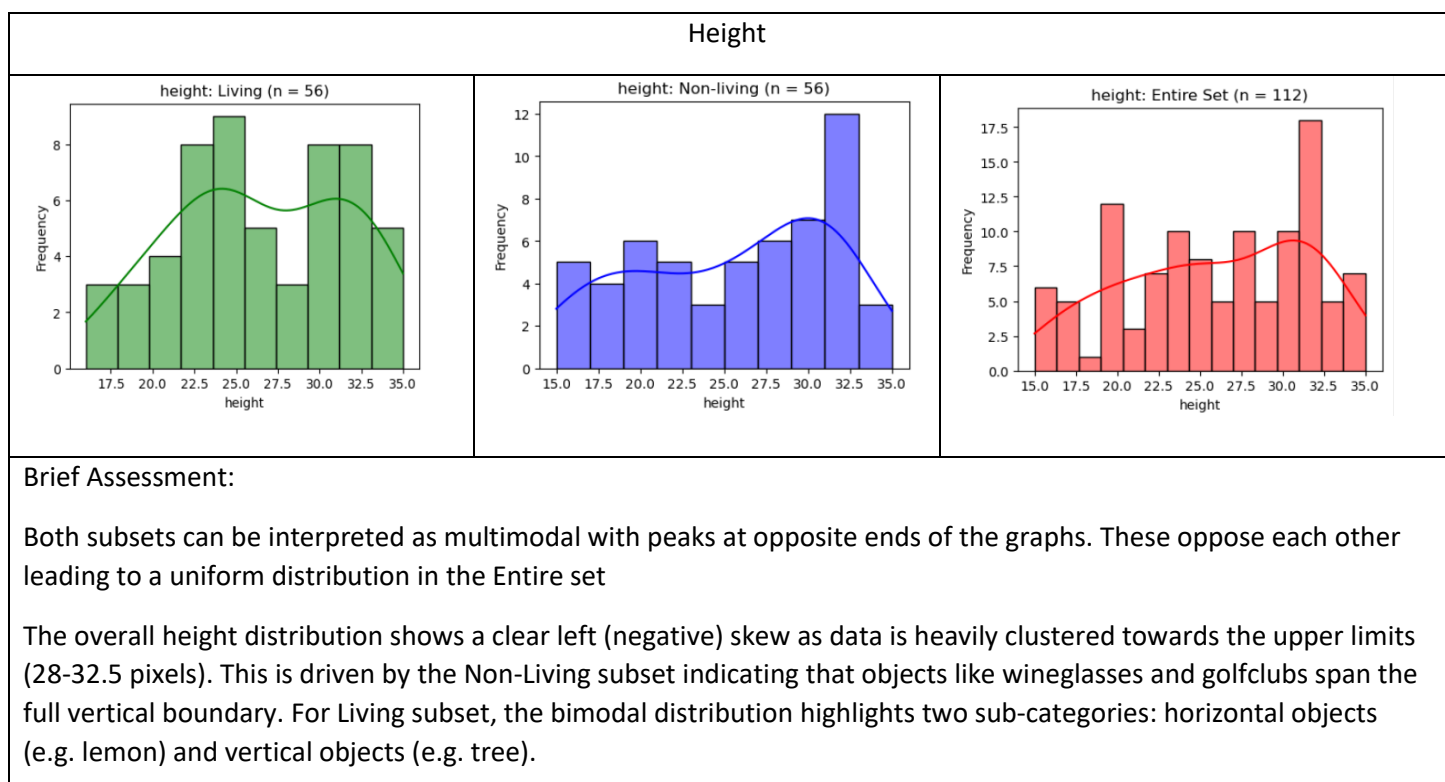
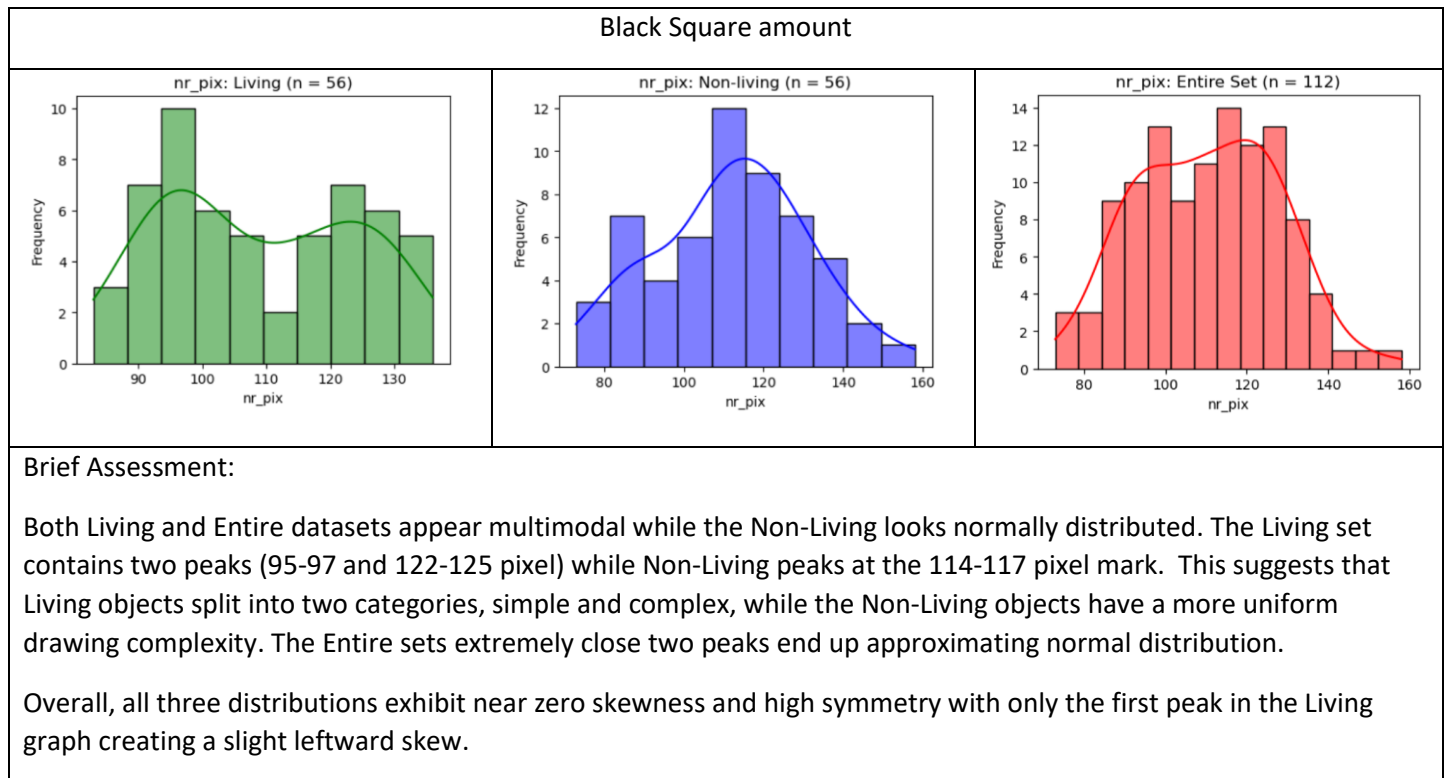
- **Hollowness**

Using the inverted 'eye' matrix and removing the first label (surrounding white area) 'sum()' was applied to gather number of white squares in the 'eyes' and then divided by the number of pixels (nr_pix)

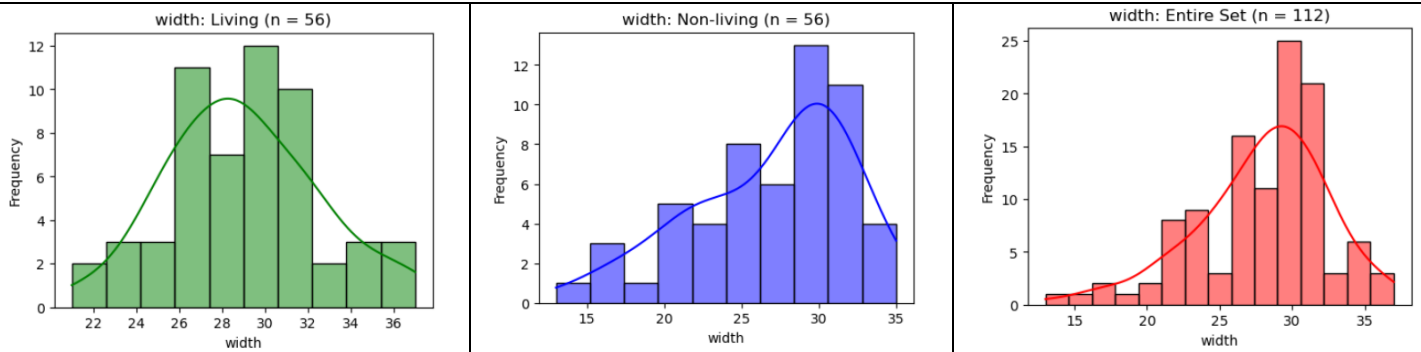
Section 3

Section 3.1

To assess the skew and normality, 12 histograms were constructed and overlaid with Kernel Density Estimate curves.



Width

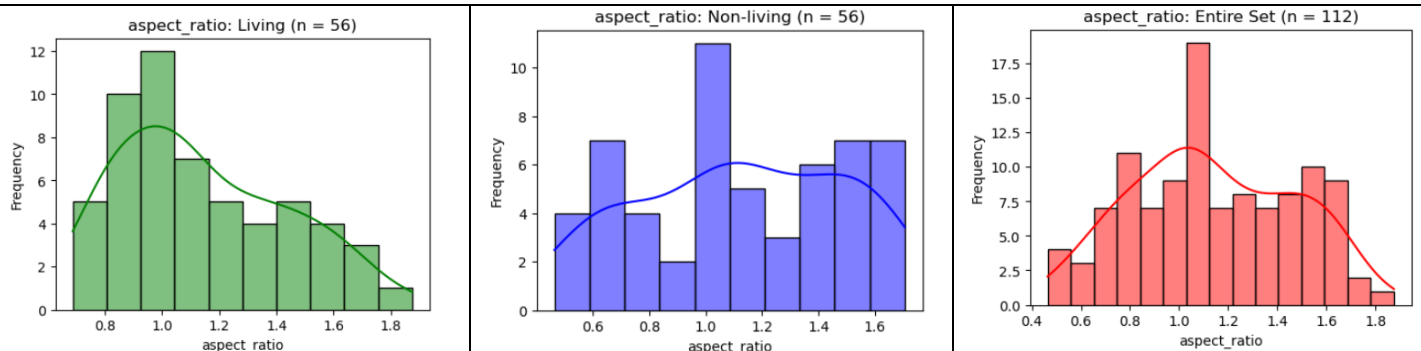


Brief Assessment:

All datasets follow a unimodal distribution characterised by a single prominent peak.

The overall distribution exhibits a strong left (negative) skew driven by the differences between the two subsets. The Non-Living subset is heavily left-skewed with data falling towards the horizontal limit (28-35 pixels). The Living subset displays a more central distribution. This suggests that Non-Living objects stretch across the maximum canvas width, compared to Living objects that are drawn in compact, round shapes (with outliers caused by longer objects like bananas).

Aspect Ratio



Brief Assessment:

The aspect ratio displays the greatest variance across subsets. The Living dataset shows a unimodal distribution that peaks early and exhibits a rightward tail driven by outliers like bananas. The Non-Living graph is highly multimodal with three peaks (lower, central and upper ranges). This corresponds to the variation of man-made objects (e.g. wineglasses). These structures, when overlapped, cause the Entire dataset to resolve to a broad, bimodal distribution.

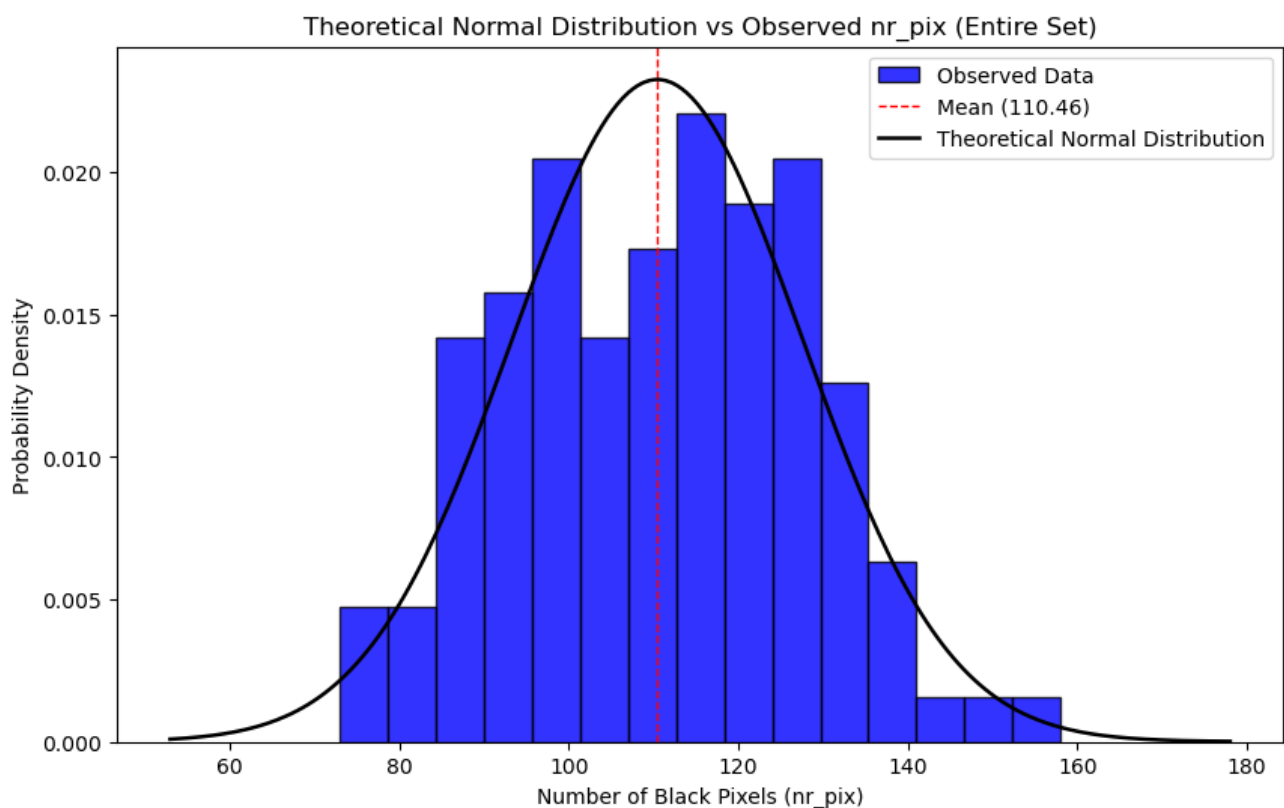
The overall distribution shows little skewness with most data clustering in the middle. The outlier to this symmetry is the Living subset that contains a moderate right (positive) skew due to its elongated tail.

Section 3.2

The sample mean ($\bar{x}$) and variance (s^2) were calculated directly from the `nr_pix` array using standard statistical functions. These parameters were used to compute the theoretical probability density function, which was then superimposed over the histogram to visually assess normality.

Calculation Results

Estimated Sample Mean ($\bar{x}$)	Estimated Sample Variance(s^2)	Estimated Sample Standard Deviation (s)
110.45	294.45	17.16



The theoretical mean of $\bar{x} = 110.46$ lands in a dip between the two clusters of data. The distribution is bimodal, featuring peaks on the left (97-102 pixel) and a taller, denser peak on the right (115-130 pixels). Due to the larger right peak, the mathematical mean is pulled rightwards and falls between the two peaks.

Consequently, the theoretical model places the highest point of its bell curve where the actual data is experiencing a dip. While data leading up to the theoretical centre remains somewhat in line with the model, the bimodal nature of the data means that the graph does not fall in line with its prediction. Ultimately the differences in the data and theory give us the conclusion that normal distribution is a poor representation of a dataset's actual most frequent pixel count.

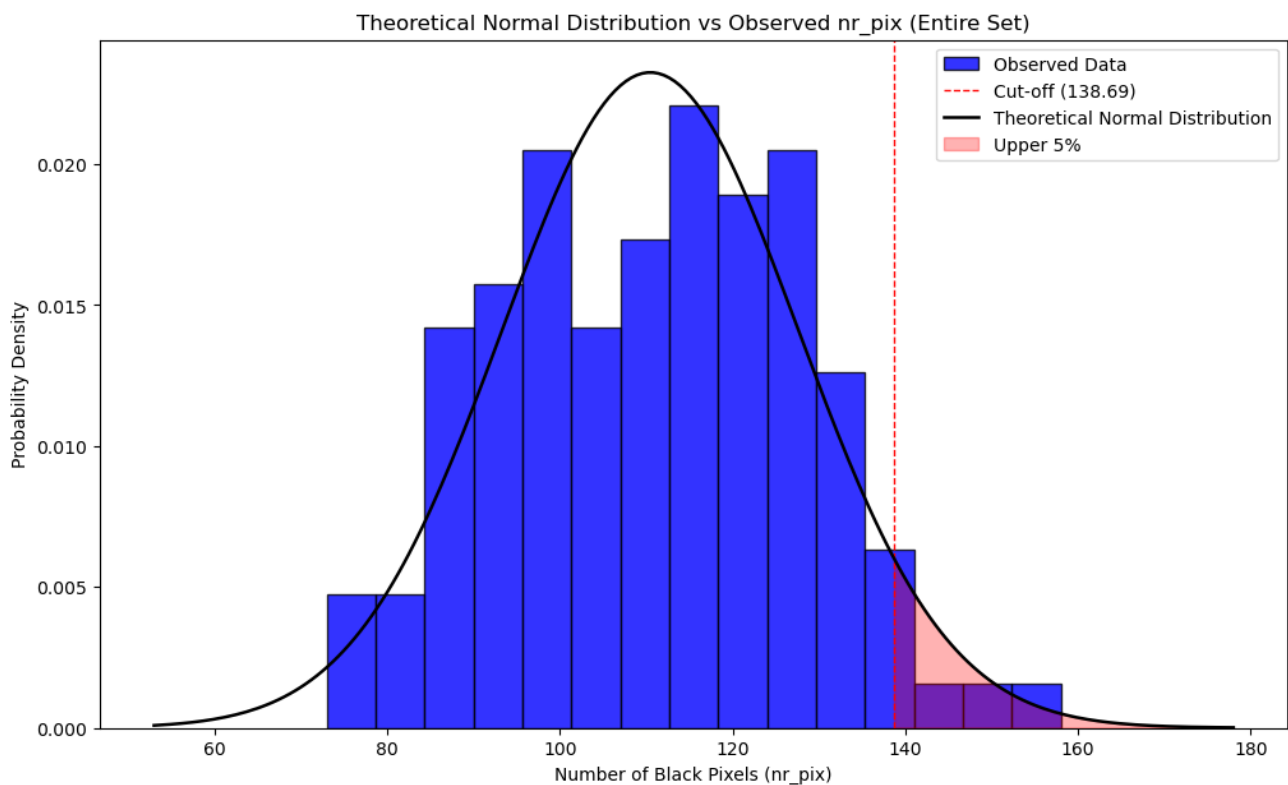
Section 3.3

Using the estimated mean ($\bar{x}$) of 110.46 and standard deviation (s) of 17.16, the `norm.ppf()` function calculated a 95% cumulative probability. By inputting a cumulative probability of 0.95, estimated mean for 'loc' and standard deviation for 'scale'. A 5% cut-off estimate of 138.69 pixels was established.

Filtering the dataset revealed only 3 instances of images over 138.69 pixels (see left figure)

The only images that fell into the Non-Living category were envelopes. These were drawn with larger bounding boxes and more complex internal features (such as a deeper "V" flap).

```
Images in top 5%:
  Object  Index  nr_pix
63  envelope    8   147.0
65  envelope   10   146.0
66  envelope   11   158.0
```



While the theoretical model predicts approximately 5.6 images should exceed this threshold, only 3 images were found to be in this range (approx. 2.7% of images). This discrepancy provides further evidence that the distribution is not perfectly normal.

Section 3.4

To determine which features were most effective in differentiating Living and Non-Living items, independent Welch's T-tests were performed across all 16 features. A significance level of alpha (α)= 0.05 was applied.

Two hypotheses were created to determine if features were effective in discrimination:

- Null Hypothesis (H_0): The average value of the feature is the same for both Living and Non-Living items.
- Alternative Hypothesis (H_1): There is a statistically significant difference between the two groups

Welch's T-Test was chosen as it does not assume equal population variance^[5], providing better results for hand-drawn data that has a spread of pixels. For T-Test, the pandas and scipy libraries were used. The following function performs that Welch test:

```
#T-test
t_test, p_val = stats.ttest_ind(living_data[f], non_living_data[f], equal_var=False)
```

Welch T-Test Formula

The above function applies this formula:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

Where $\bar{x}_1$ and $\bar{x}_2$ are Living and Non-Living groups, s_1^2 and s_2^2 are the variance of each group and n_1 and n_2 are sample sizes

Welch's T-Test

Features were considered significant if their T-statistic exceeded the threshold of $|1.98|$ standard errors. The T statistic being greater than indicates that the difference is not due to random chance (probability < 5%).

P-Value

To confirm these findings, the P-values were analysed for significance. Any features with a P-value less than 0.05 allow the rejection of the null hypothesis with 95% confidence.

Welch's T-Test Results	
Features	Results
Nr_pix	-0.980005
Rows_with_1	-2.329434
Cols_with_1	3.592354
Rows_with_2	0.852757
Cols_with_2	2.096076
Rows_with_3p	1.122274
Cols_with_3p	-1.070989
Height	0.969929
Width	2.506279
Aspect_ratio	0.264749
Maxrow	-2.966986
Maxcol	-4.433397
Connected_areas	-0.792916
Eyes	-3.964125
Hollowness	4.235063
Extent	-3.711330

The P-values for the tests were obtained using the scipy library which maps the t-statistics from the Welch's test to a Student's t-distribution (due to small sample size). The function adjusts the degrees of freedom to account for the unequal variances observed between the subsets, allowing us to obtain credible P-value's that determine statistical significance.

P value and Null Hypothesis Results		
Features	P - Value	Result
Maxcol	0.000034	Reject Null Hypothesis (H_0)
Hollowness	0.000051	Reject Null Hypothesis (H_0)
Eyes	0.000155	Reject Null Hypothesis (H_0)
Extent	0.000397	Reject Null Hypothesis (H_0)
Cols_with_1	0.000594	Reject Null Hypothesis (H_0)
Maxrow	0.003863	Reject Null Hypothesis (H_0)
Width	0.013854	Reject Null Hypothesis (H_0)
Rows_with_1	0.022338	Reject Null Hypothesis (H_0)
Cols_with_2	0.038373	Reject Null Hypothesis (H_0)
Rows_with_3p	0.265236	Fail to Reject Null Hypothesis (H_0)
Cols_with_3p	0.287116	Fail to Reject Null Hypothesis (H_0)
Nr_pix	0.329334	Fail to Reject Null Hypothesis (H_0)
Height	0.334235	Fail to Reject Null Hypothesis (H_0)
Rows_with_2	0.395647	Fail to Reject Null Hypothesis (H_0)
Connected_areas	0.429898	Fail to Reject Null Hypothesis (H_0)
Aspect_ratio	0.791720	Fail to Reject Null Hypothesis (H_0)

P-Value Assessment

Nine features were found to be statistically significant. The strongest discriminators (e.g. maxcol, hollowness, eyes and extent) suggest that the largest differences between the Living and Non-Living groups is their internal complexity and structural density. While all features are significant, the rest have less of an impact when it comes to differentiating the two categories.

Interpretation of results

Significant differentiating features

The strongest evidence of discrimination points to a difference in the shapes are constructed

Hollowness and Eyes: Living objects show higher 'hollowness' (mean = 2.24) compared to Non-Living (mean = 1.15) suggesting that objects like cherries and lemons are drawn more spread out. Conversely, Non-Living items have a higher 'eye' count due to the more complicated internal structure of Non-Living object (e.g. pencil, envelopes).

Maxcol: Non-Living items showed a higher 'maxcol' (12.38 vs 9.02), indicating that items like the golfclub and pencil utilize more horizontal space or display more extreme pixel placement.

Extent: The custom metric backs up the disparity in maxcol mean. Non-Living objects fill 17.8% of their bounding boxes while a Living item only fills about 14.6% on average. Non-Living objects tends to fill their bounding box more efficiently leading to more complex internal bound boxes. Living objects tend to have more white space within their boundaries due to their curved edges and stems that don't conform as neatly to a square or rectangle.

Non-Significant differentiating features

Seven features showed no statistical significance in difference, remaining consistent across the dataset.

Nr_pix: With a 0.329 P-value, Living objects (Mean = 108.9) use a similar amount of pixels as Non-Living objects (Mean = 112.1). This suggests I tended to use the same amount of 'ink' per image.

Connected_areas: The P-value of 0.430 suggest that during the drawing of both subsets I tended to draw single, continuous strokes leading to a very small number of disconnected parts.

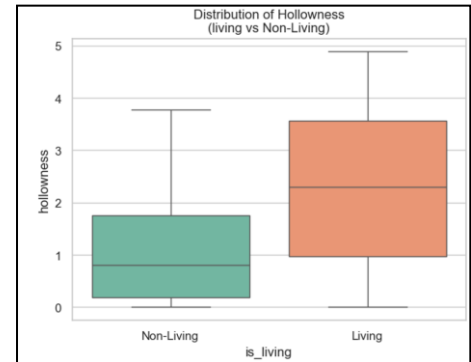
Aspect_ratio: The least discriminating feature with a P-value of 0.792. Living and Non-Living subsets were almost identical in mean (1.14 vs 1.13). This revealed that the overall boxiness of the drawings were near identical across all categories.

The failure of these features to discriminate suggest while the internal complex features like hollowness and eyes varies, the external shell of the drawings remain uniform across the dataset.

Visual Illustrations

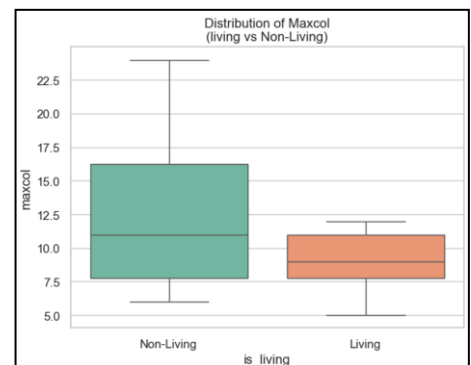
Hollowness

Living box sits significantly higher on Y-axis than Non-Living box as Living objects were drawn with more internal white space. The gap between the interquartile ranges (IQRs) provides visual confirmation of the highly significant P-value ($P=0.000051$). Furthermore, the median line for the Living category is nearly double that of the Non-Living category (2.24 vs 1.15) and the wider IQR in the Living group highlights a greater variability in how the Living group textures were rendered compared to the more uniform and solid structures of the Non-Living group.



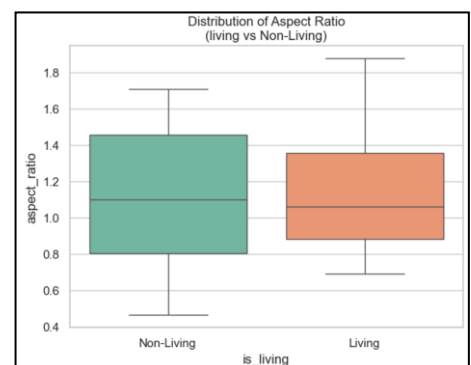
Maxcol

The maxcol plot shows a difference in horizontal reach, with the Non-Living IQR sitting notably higher (12.38 vs 9.02). The difference in reach and box size provides visual confirmation to the significant P-value ($P=0.000034$). The wider variability in the Non-Living group reflects the structural gaps between elongated and compact objects.



Aspect Ratio

This boxplot illustrates a lack of discrimination, with both IQRs overlapping almost entirely. Nearly identical medians (1.14 vs 1.13) confirm the high P-value ($P=0.792$). These results confirm the earlier conclusion that while the internal complex features like hollowness and eyes varies, the external shell of the drawings remain uniform across the dataset.



Section 3.5

Pearson's correlation coefficient^[5] was deemed the most appropriate choice for investigating the relationship between the features (height, width, aspect ratio, hollowness and number of pixels). The method was selected because the variables are quantitative and the visual assessments in Section 3.1 suggest the distributions, while sometimes being multimodal, are sufficiently normal to satisfy the assumptions of the test.

Using the `corr()` function from the Pandas library I applied it to the subset of the five specific features. The `corr()` function included `method='pearson'` to make sure that the Pearson's correlation coefficient was applied.

Results

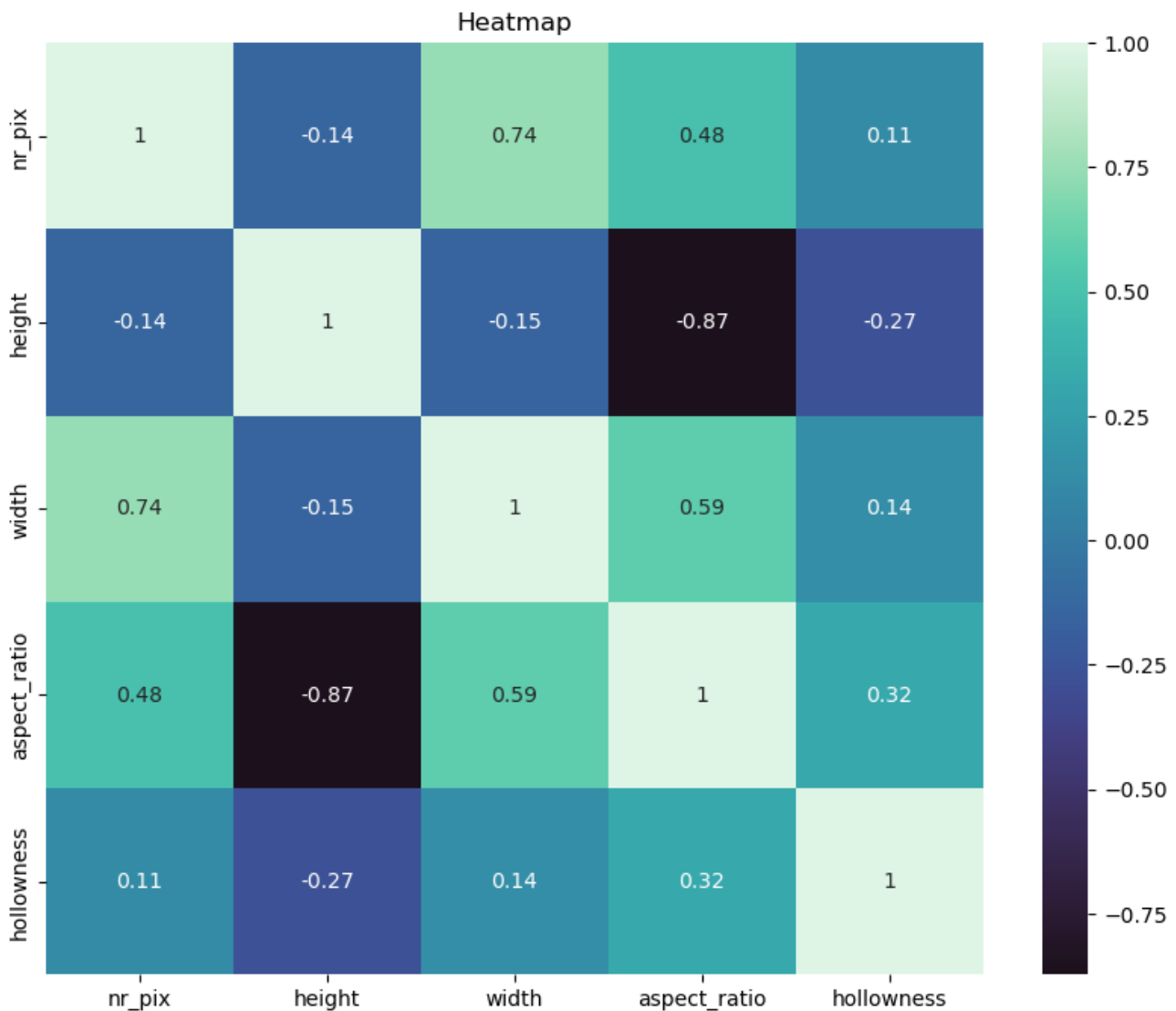
Features	Nr_pix	Height	Width	Aspect Ratio	Hollowness
Nr_pix	1.000000	-0.140591	0.737550	0.481540	0.109290
Height	-0.140591	1.000000	-0.148336	-0.871997	-0.272333
Width	0.737550	-0.148336	1.000000	0.586768	0.136306
Aspect Ratio	0.481540	-0.871997	0.586768	1.000000	0.317272
Hollowness	0.109290	-0.272333	0.136306	0.317272	1.000000

Features that show a positive correlation ($r > 0$) indicate that when one feature's value increases the other feature's value follows. In the data we can see that the features like `nr_pix` and `width` ($r = 0.738$), and `aspect ratio` and `width` ($r = 0.587$) have a strong positive correlation.

Features showing a negative correlation ($r < 0$) indicate that when one feature's value increases, the others value decreases. `Height` and `Aspect ratio` ($r = -0.872$) is the only strong negative correlation present in this data.

Features closer to zero like `Nr_pix` and `Hollowness` (0.109), and `Nr_pix` and `Height` (-0.141) show little to no linear association.

Heat Map



The heatmap puts these results into a visual illustration.

The results from these tests point towards a few suggestions.

- The positive correlation between Nr_pix and Width ($r = 0.738$) indicate that wider drawings naturally require more pixels. On the contrary, Nr_pix and Height ($r = -0.141$) show a lack of correlation, suggesting that for this dataset, pixel count was primarily driven by the horizontal complexity of the object rather than its vertical span. It indicates that even as objects grew taller (e.g. tree), their ink density did not increase proportionally in the same way it did when the object became wider.
- The Negative correlation between Height and Aspect ratio ($r = -0.872$) is the strongest association in the dataset. This makes sense however due to the feature's definition ($AR = W/H$). Aspect ratio value decreases as Height increases in the calculation, so correlation is understandable.
- Values close to zero such as Hollowness and Nr_pix ($r = 0.109$) indicate a weak linear association, suggesting that the complexity of an object's internal structure do not affect the overall pixel count in the dataset.

Statistical Inference

Statistical inference allows us to determine if the patterns found by the Pearson's correlation coefficient are likely to exist in the wider population. It will help distinguish between a genuine structural trend and a result of random variation in my drawing style.

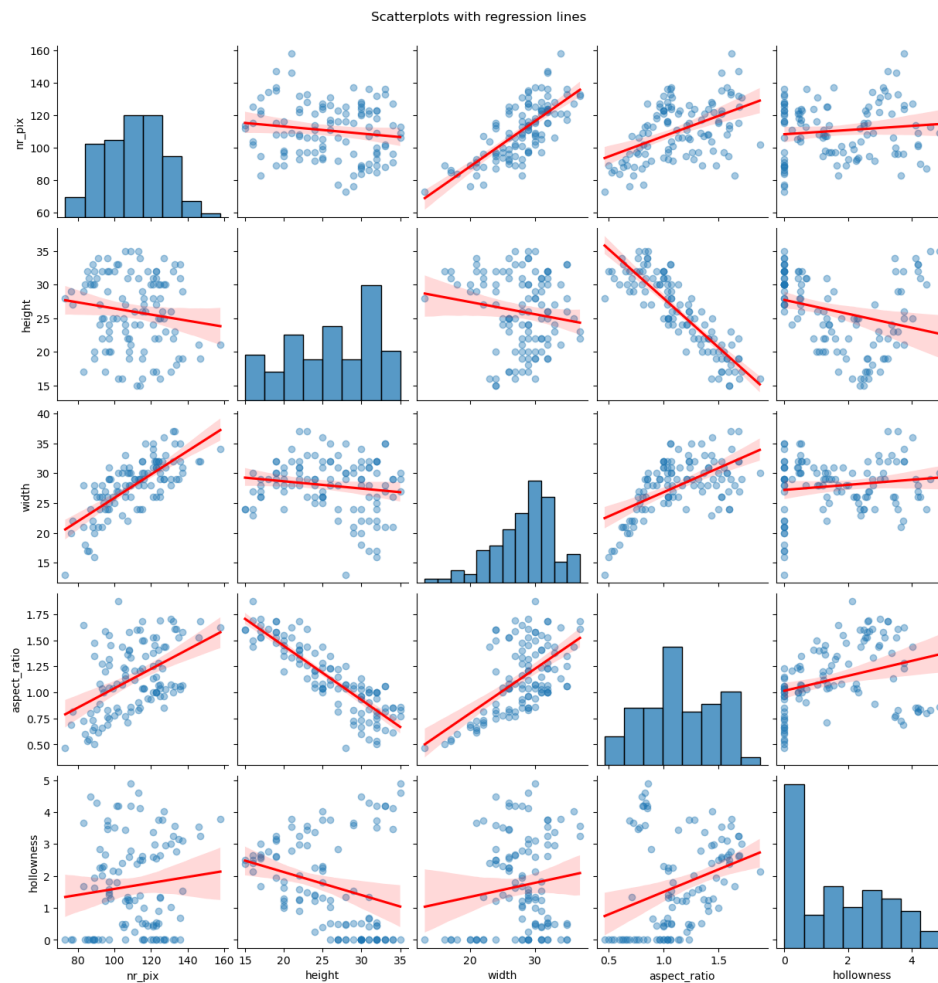
Two hypotheses were created to determine if a statistically significant linear relationship exists between the two pairs

- Null Hypothesis (H_0): There is no linear relationship between the two features.
- Alternative Hypothesis (H_1): There is a statistically significant linear relationship between the two features.

To determine if the linear association between features were significant, P-values were obtained using the `stats.pearsonr` function from the `scipy` library. The function calculates the Pearson Correlation Coefficient and transforms this value into a t-statistic which is compared against a Student's t-distribution. This allows us to test the Null Hypothesis H_0 . A significance level of alpha (α)= 0.05 was applied. Any feature pair with a P-value less than 0.05 indicates a correlation highly unlikely to have occurred randomly allowing us to reject the Null Hypothesis with 95% confidence.

Inference for Pearson Correlation Coefficient		
Features	P-Value	Results
Nr_pix and Height	0.1393	Fail to Reject Null Hypothesis (H_0)
Nr_pix and Width	0.0000	Reject Null Hypothesis (H_0)
Nr_pix and Aspect Ratio	0.0000	Reject Null Hypothesis (H_0)
Nr_pix and Hollowness	0.2513	Fail to Reject Null Hypothesis (H_0)
Height and Width	0.1186	Fail to Reject Null Hypothesis (H_0)
Height and Aspect Ratio	0.0000	Reject Null Hypothesis (H_0)
Height and Hollowness	0.0037	Reject Null Hypothesis (H_0)
Width and aspect ratio	0.0000	Reject Null Hypothesis (H_0)
Width and Hollowness	0.1518	Fail to Reject Null Hypothesis (H_0)
Aspect Ratio and Hollowness	0.0007	Reject Null Hypothesis (H_0)

Scatterplot with regression lines



Interpretation of Results

The scatterplot matrix provides a visual illustration that correlates with the “Reject H_0 ” results from the inference table. By observing the distribution of the data points along the regression lines we can draw the following conclusions:

- **Height and Aspect ratio:** The plot displays a tight, linear cluster of data points with a steep downwards slope, confirming the negative correlation ($r = -0.872$). This and the highly significant P-value ($P = 0.00001$), confirm a strong dependency. As height increase, the Aspect ratio must mathematically decrease, assuming width stays the same or increases at a slower rate.
- **Nr_pix and Width:** A distinct upwards trend is revealed in the scatterplot closely around the regression line ($r = 0.738$). The extremely low P-value ($P = 0.0000$) confirms that across a population, the wider an object was, the more pixels would be used. Objects like envelopes and pencils naturally required more pixels compared to a circular cherry.
- **Non-Significant relationships:** Plots such as Nr_pix and Height and Nr_pix and Hollowness show a largely dispersed set of data points with near flat regression lines. These visuals support the “Fail to Reject H_0 ” as both P-values were greater than 0.05 proving that vertical height and internal complexity did not have a reliable influence on total pixel count.

Section 3 Discussion

The statistical analysis of the 112 hand-drawn images provides a clear profile of the dataset's characteristics.

Feature and class discrimination

The Welch's T-Test and P-value analysis successfully identified nine features that act as reliable indicators for distinguishing between Living and Non-Living objects.

- **Best Indicators:** hollowness and maxcol are the most effective discriminators. The significantly higher hollowness in Living objects (2.24 vs 1.15) reflects a drawing style that prioritises white space, while the higher maxcol in Non-Living objects (12.38 vs 9.02) highlights the more structured and space-filled nature of manmade objects.
- **Non-Significant features:** Seven features, most notably aspect ratio ($P = 0.792$) and nr_pix ($P = 0.329$), failed to show any statistical difference. This suggests that the number of pixels and external box of the drawings were independent across the entire dataset.

Structure and Linear Association

Pearson's Correlation analysis confirmed that many features were structurally linked rather than category driven.

- The strong negative correlation between height and aspect ratio proves that the shape's proportions are a direct mathematical result of its dimensions.
- The strong association between width and nr_pix suggest that as objects were drawn wider, they naturally contained more pixels.

The analysis concludes that while the external parts of my drawings (aspect ratio, total pixels) remain consistent across the dataset, the internal complexity (hollowness, eyes) provides a significant indicator for telling the two subsets apart.

Section 4

Section 4.1

To predict the aspect_ratio, a Multiple Linear Regression (Ordinary Least Squares) model was selected. This model is appropriate because the target variable (aspect_ratio) is a continuous quantitative feature and the goal is to map the linear relationship between it and other independent continuous features.

To predict the aspect ratio as parsimoniously as possible, height and width were chosen as sole predictors. This is due to statistical findings from Section 3.5:

- **Statistical Evidence:** Pearson's correlation tests revealed that height ($r = -0.872$) and width ($r = 0.587$) had the strongest linear association with aspect ratio.
- **Dependency:** Aspect ratio is mathematically derived from these two features.
- **Efficient:** Including other features would introduce unnecessary complexity and risk overfitting as they do not dictate the dimensions of the object.

Section 4.2

The OLS regression model was fitted using the statsmodels library. The resulting metrics confirm the model's high accuracy and statistical significance.

Metric Assessment

- R-squared and Adjusted R-squared: The model produced an R-squared of 0.9743, meaning that height and width explain 97.43% of aspect ratio variance. The near identical Adjusted R-squared (0.9738) confirms model parsimony.
- Mean Squared Error (MSE): The extremely low MSE (0.0028) indicated that the average squared difference between the observed and predicted aspect ratio is negligible, showing a high predictive accuracy.

Metric	Value
R-Squared	0.9743
Adjusted R-squared	0.9738
Mean Squared Error	0.0028
Intercept	1.4287
Coefficient Height	-0.0477
Coefficient Width	0.0339

The linear model equation

$$y = b_0 + b_1x_1 + b_2x_2$$

Using the Linear model equation(above figure), the predictive equation for this model is:

$$\text{Aspect Ratio} = 1.4287 - 0.0477(\text{Height}) + 0.0339(\text{Width})$$

This equation confirms the negative relationship found in Section 3. As height increases, the predicted aspect ratio decreases. In contrast, an increase in width increases the ratio.

Section 4.3

Guided by the results in Section 3.4, maxcol was selected as the most discriminating feature due to it containing the lowest p-value (0.000034). Additionally, that Non-Living objects consistently showed a higher horizontal reach (mean = 12.38) compared to Living objects (mean = 9.02).

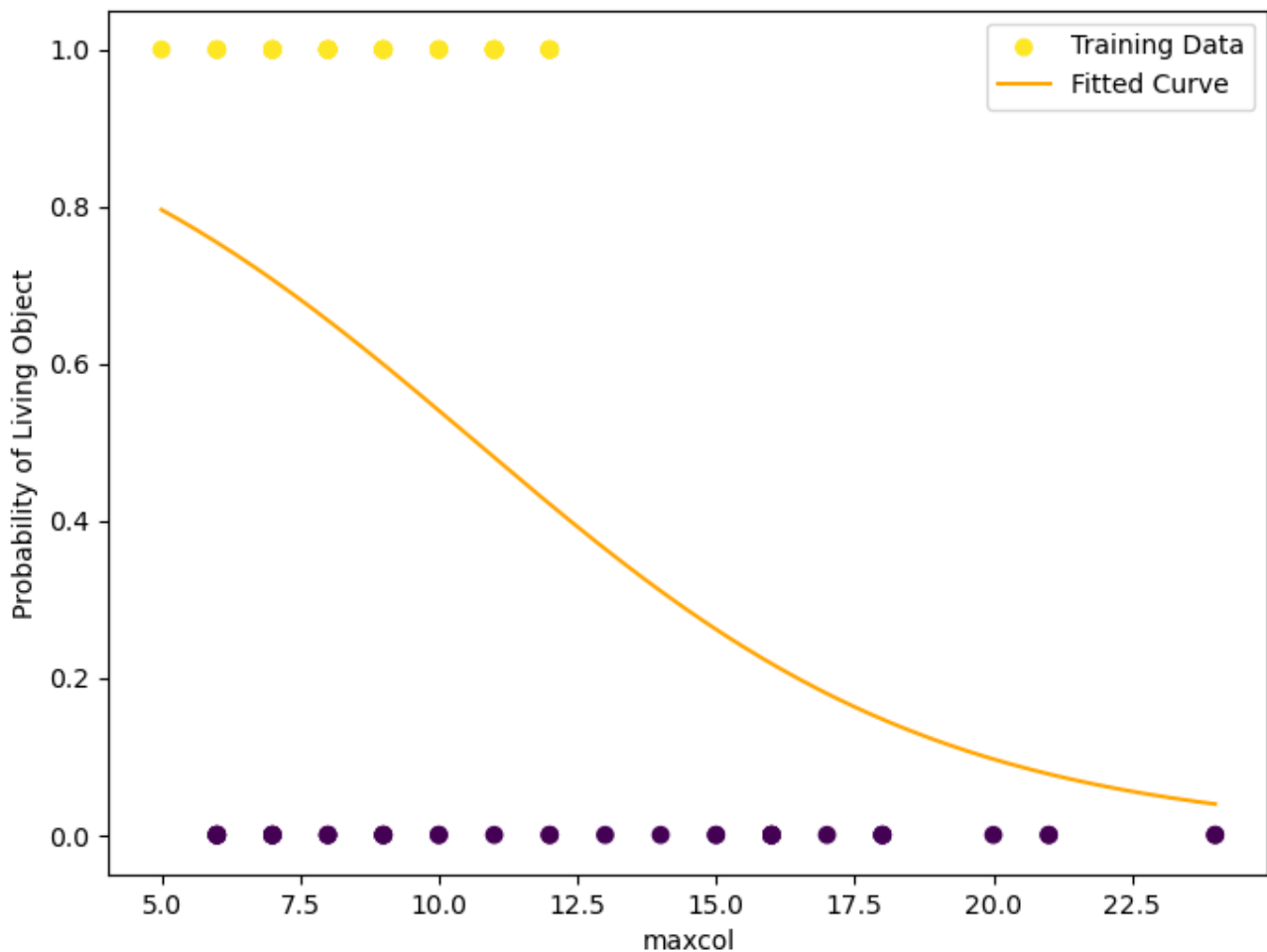
To perform the classification, a Logistic Regression model was implemented using the scikit-learn library. The dataset was shuffled and split into an 80% training set and a 20% testing set. A binary target variable was created, assigning 1 to Living objects and 0 to Non-Living objects.

Model Results

The model was fitted to the training data and calculated the following parameters (see figure left).

The negative coefficient (-0.2399) confirms the negative relationship identified in Section 3. As the maximum column width of a drawing increases, the probability of it belonging to the Living category decreases.

Data	Result
Intercept	2.5611
Coefficient for Maxcol	-0.2399
Training Accuracy	62.92%
Testing Accuracy	56.52%



Visual Illustration

The scatterplot above maps the binary training data, where 0 = Non-Living and 1 = Living, against maxcol. The downwards-sloping curve visually illustrates that drawings with a low maxcol have a higher probability (>0.5) of being classified as Living. Those with an extended horizontal reach fall towards a probability of 0, classifying them as Non-Living.

While the model can correctly identify the structural trend, the moderate accuracy scores (Train = 63%, Test = 56%) and the overlap of data points in the centre of the graph indicate that by itself, maxcol is not entirely sufficient to achieve perfect classification across the dataset.

Section 4 Interpretation of results

The regression analyses demonstrate two distinct levels of predictive success. The multiple linear regression model for aspect_ratio achieved near-perfect accuracy with an R-squared of 0.9743, confirming that the external dimensions are controlled by the features mathematical dependencies.

In contrast, the logistic regression model for category classification using maxcol showed moderate results with a training accuracy of 63% and testing accuracy of 56%. This indicates that while an object's horizontal reach is a significant discriminator, dimensional overlaps between the categories (such as elongated Living objects like bananas sharing similar max column width with Non-Living objects like pencils) require a more complex approach that includes multiple features for reliable classification.

Conclusion

This report successfully details the creation, processing and analysis of a custom dataset of 112 hand-drawn images.

Following the extraction of 16 features within the dataset, statistical analyses were performed. Welch's T-Test and Pearson's Correlation Coefficient revealed significant structural patterns leading to a fundamental conclusion. While the external boundaries of the drawings (such as total number of pixels) remained uniform across the dataset, the internal structural complexity provided the strongest discriminator of categories. Maxcol and Hollowness were found to be the strongest indicators of this internal complexity.

These findings directly informed the subsequent machine learning models. A parsimonious multiple linear regression model successfully mapped the mathematical dependency of aspect ratio on height and width, achieving a high predictive accuracy (R-squared = 0.9743). In contrast, the logistic regression model utilizing maxcol provided a moderate testing accuracy of 56.52% for binary classification. This moderate performance highlights the limitation of using a single feature, as the dimensional overlap between the objects restricts its predictive power.

Overall, this investigation demonstrates that while individual features can be powerful discriminators, the varying complexity of hand-drawn objects require an approach containing more features, rather than relying on a singular indicator.

References

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AI Usage Declaration

AI was used to help understand lecture notes more clearly, debugging and provide graph colour palettes. All data extraction and analysis did not use AI